

⇒ Laplace transform: The word Laplace means 3 things. 1, Integral 2, Exponential 3, factors
→ it is denoted by L &

General form:- $L\{f(t)\} = \int_0^{\infty} f(t) e^{-st}$

→ Prove $Laplace\{1\} = \frac{1}{s}$

Sol

$$L\{1\} = \int_0^{\infty} e^{-st} \cdot f(s)$$

$$L\{1\} = \int_0^{\infty} e^{-s(t)} \cdot (1)$$

$$L\{1\} = \int_0^{\infty} e^{-s}$$

Now Applying limit

$$L\{1\} = \left[e^{-s(\infty)} - \frac{e^{-s(0)}}{-s} \right]$$

$$L\{1\} = e^{-\infty} - e^0$$

$$L\{1\} = \frac{0}{-s} - \frac{1}{-s}$$

$$\boxed{L\{1\} = \frac{1}{s}} \text{ Proved.}$$

Q5, Solve the differential eq subject to the indicated to the initial cond: by Laplace transform.

$$y'' - 4y' + 4y = (12x^2 - 6x)e^{2x}$$

⇒ First shifting theorems / Property

$$\mathcal{L}\{P(t)\} = \mathcal{L}\{P(s)\}$$

ex:-

$$\mathcal{L}\{e^{at} \cdot \sin 3t\}$$

$$\Rightarrow \boxed{\mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}}$$

$$\mathcal{L}\{e^{2t}\} \cdot \mathcal{L}\{\sin 3t\}$$

$$\text{note: } \mathcal{L}\{e^{2t} \cdot e^{3t}\}$$

$$\mathcal{L}\left\{\frac{a}{s-a}\right\}$$

$$\mathcal{L}\left\{\frac{3}{(s-2)^2 + (3)^2}\right\}$$

$$\frac{1}{(s-2)} \cdot \frac{1}{(s-3)}$$

$$e^{st} = \frac{1}{(s-s)}$$

$$\mathcal{L}\left\{\frac{3}{(s-2)^2 + (3)^2}\right\}$$

ex 3

$$\mathcal{L} \{ t^n \cdot f(t) \} = (-1)^n \frac{d^n}{ds^n} [F(s)]$$

ex:

$$\mathcal{L} \{ t^2 \cos t \} = (-1)^2 \frac{d}{ds} \frac{s}{s^2 + a^2}$$

$$\mathcal{L} \{ \cos t \} = \frac{1}{s^2 + a^2}$$

$$\mathcal{L} \{ t^2 \cdot \cos t \} = (-1)^2 \frac{d}{ds} \frac{s}{s^2 + a^2}$$

(1) $\times \frac{d}{ds} \frac{s}{s^2+1} \rightarrow$ b) using the Quotient rule of derivative

Let $u = s$ $v = s^2 + 1$

$$\boxed{\frac{v u' - u v'}{v^2}}$$

or

$$\frac{(s^2+1) \frac{d}{ds}(s) - [(s) \cdot 2s]}{(s^2+1)^2}$$

$$\frac{(s^2+1) \cdot 1 - 2s^2}{(s^2+1)^2}$$

$$\frac{(s^2+1) - 2s^2}{(s^2+1)^2} \rightarrow \frac{s^2+1-2s^2}{(s^2+1)^2} \rightarrow \boxed{\frac{1-s^2}{(s^2+1)^2}} \text{ first derivative}$$

so now for second derivative again using
Quotient rule of derivative

$$\rightarrow \frac{1-s^2}{(s^2+1)^2} \quad \text{Let } u = 1-s^2 \\ v = (s^2+1)^2$$

$$\frac{v u' - u v'}{v^2} \\ (s^2+1)^2 - 2s - [(1-s^2) 2(s^2+1) \cdot 2s]$$

$$\frac{(s^2+1)^4 - 2(s^2+2s+1) - [(1-s^2) 4s^3 - 4s]}{(s^2+1)^4} \quad \text{or}$$

$$\frac{-2s(s^2+1)^2 - [(1-s^2) 4s(s^2+1)]}{(s^2+1)^4}$$

$$\frac{(s^2+1)^4 [-2s(s^2+1) - [(1+s^2) 4s]]}{(s^2+1)^4}$$

$$\frac{-2s^3 + 2s - 4s - 4s^3}{(s^2+1)^3}$$

$$\frac{2s^3 - 6s}{(s^2+1)^3} \quad \text{Ans.}$$

Note: $\log \infty = \infty$

⇒ some practice Question of Laplace transform-

ex 1,

$$\mathcal{L} \left\{ \frac{e^{-4t} - e^{2t}}{t} \right\}$$

ex 2

$$\mathcal{L} \left\{ \frac{1 - \cos 3t}{t} \right\}$$

ex 1, $\mathcal{L} \left\{ \frac{e^{-4t} - e^{2t}}{t} \right\}$

sol

$$\mathcal{L} \left\{ e^{-4t} - e^{2t} \right\} = \mathcal{L} \left\{ e^{-4t} \right\} - \mathcal{L} \left\{ e^{2t} \right\} = \frac{1}{(s+4)} - \frac{1}{s-2}$$

"using linearity property"

Now we have to find.

$$\mathcal{L} \left\{ \frac{e^{-4t} - e^{2t}}{t} \right\} \text{ so for that we have to Apply integral}$$

$$\int_0^{\infty} \left[\frac{1}{(s+4)} - \frac{1}{(s-2)} \right] ds = \int_0^{\infty} \left[\frac{1}{(s+4)} - \frac{1}{(s-2)} \right] ds$$

so in denominator and in Numerator its derivative is present so for that

$$\left[\log [s+4] - \log [s-2] \right]_0^{\infty}$$

$\left[\log [s+4] - \log [s-2] \right]_0^{\infty}$ so when we put limit ∞ the $\log \infty$ is ∞ which is not define.

$$\left[\log \frac{(s+4)}{(s-2)} \right]$$

$$\log \frac{\left[1 + \frac{4}{s} \right]}{\left[1 - \frac{2}{s} \right]} \Rightarrow s \text{ common by force}$$

$$\frac{a}{\infty} = 0$$

$$\log 1 = 0$$

Now Applying Limit
as upper limit - lower
limit.

$$= \log \left[\frac{(1 + \frac{a}{s})}{(1 + \frac{b}{s})} - \log \frac{(1 + \frac{a}{s})}{1 + \frac{a}{s}} \right]$$

$$= \log \left[\frac{(1+0)}{(1+0)} - \log \frac{(s+a)}{(\frac{s+b}{s})} \right]$$

$$= \log 1 - \log \frac{(s+a)}{s+b}$$

$$= 0 - \log \frac{(s+a)}{(s+b)}$$

$$= \log \frac{(s+b)}{(s+a)}$$

So

$$\lim_{t \rightarrow \infty} \left[\frac{e^{-4t} - e^{2t}}{t} \right] = \log \frac{(s+b)}{(s+a)}$$

Ans

ex: $\mathcal{L} \left\{ \frac{1 - \cos 3t}{t} \right\}$

Sol

First find the $\mathcal{L}\{1 - \cos 3t\}$

$\mathcal{L}\{1\} - \mathcal{L}\{\cos 3t\}$ by using linearity property &

$$\frac{1}{s} - \frac{s}{s^2 + 3^2}$$

$$\frac{1}{s} - \frac{s}{s^2 + 9}$$

now find the

$$\mathcal{L} \left\{ \frac{1 - \cos 3t}{t} \right\} =$$

$$\int_s^\infty \left(\frac{1}{u} - \frac{u}{u^2 + 9} \right) du$$

$$\int_s^\infty \left(\log(u) - \frac{1}{2} \times \frac{du}{u} \right)$$

$$\int_s^\infty \left(\log(u) - \frac{1}{2} \log u \right)$$

$$\int_s^\infty \left(\log(u) - \log(u)^{1/2} \right)$$

$$\log \left[\frac{u}{(u^2 + 9)^{1/2}} \right]_s^\infty$$

$$\log \left[\frac{u}{(u^2 + 9)^{1/2}} \right]_s^\infty$$

$$\text{let } u = s^2 + 9$$

$$\frac{du}{ds} = 2s$$

$$du = 2s ds$$

$$\frac{du}{2} = s ds$$

$$C_1 = 0,$$

$$\ln \frac{1}{1+\frac{9}{s^2}} = \ln \left(\frac{1}{1+\frac{9}{s^2}} \right)$$

$$\ln 1 - \ln \frac{9}{(s^2+9)^{\frac{1}{2}}}$$

$$= - \ln \left(\frac{9}{s^2+9} \right)^{\frac{1}{2}}$$

$$= -\frac{1}{2} \ln \left(\frac{9}{s^2+9} \right)$$

$$= \frac{1}{2} \left(\frac{s^2}{s^2+9} \right)^{-1}$$

$$= \frac{1}{2} \left(\frac{s^2+9}{s^2} \right)$$

so

$$\left\{ \frac{1-\cos 3t}{t} \right\} = \int_0^\infty \frac{1}{2} \left(\frac{s^2+9}{s^2} \right) e^{-st} ds$$